25901752 **A**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCPHY100 - OBSERVATIONAL ASTRONOMY

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. According to the Copernican model, which celestial body is at the center of the universe? [U] / [CO1]

   a). Jupiter b). Mars c). Sun d). Earth

2. Which unit is larger: 1 light year or 1 Astronomical Unit (AU) or 1 parsec ? [U] / [CO1]

   a). 1 light year b). 1 AU c). 1 parsec d). All are equal

3. What force keeps planets in orbit around the Sun? [U] / [CO1]

   a). Magnetism b). Gravity c). Friction d). Electricity

4. Which of the following constellations is known as "The Hunter"? [K] / [CO4]

   a). Leo b). Orion

c). Taurus d). Ursa Major

5. During which season do many trees shed their leaves? [K] / [CO4]

a). Summer b). Winter

c). Autumn d). Spring

6. What kind of rays are used in radio astronomy? [K] / [CO2]

a). Microwaves b). X-Rays

c). Radio Waves d). Visible Light

7. Which Space agency is monitoring James Webb Space Telescope? [K] / [CO2]

a). NASA b). ISRO c). ESA d). SPACE X

8. What do refractive telescopes use to focus light? [K] / [CO2]

a). Mirrors b). Lenses c). Prisms d). Filters

9. What is the approximate surface temperature of Sun? [K] / [CO5]

a). 25,000 K b). 10,000 K c). 6,000 K d). 3,000 K

10. What is the approximate duration of totality in a solar eclipse ? [K] / [CO5]

a). 2 mins b). 7.5 mins

c). 15 mins d). 30 mins

11. Which among the following is not a Jovian planet? [K] / [CO5]

a). Mercury b). Neptune

c). Jupiter d). Saturn

12. Which type of galaxy has well-defined arms? [K] / [CO4]

a). Elliptical b). Spiral

c). Irregular d). None

13. The birth stage of a star is named as: [K] / [CO3]

a). Netron Star b). Supernova

c). Protostar d). Red Giant

14. Which spectral class is the Sun classified under? [K] / [CO3]

a). F b). G c). K d). M

15. Compared to a main sequence star, a red giant is: [U] / [CO3]

a). Smaller and Cooler b). Larger and Cooler

c). Smaller and Hotter d). Larger and Hotter

[1x15 = 15]

Part B

Multiple Choice Questions

Answer all questions

Each question carries **2** marks

16. What does the term "deferent" refer to in the Ptolemaic model? [K] / [CO1]

a). The small circular orbit of a planet

b). The retrograde motion of planets

c). The larger circle along which the center of the epicycle moves

d). The center of the universe

17. Which book did Galileo publish to support the heliocentric theory? [K] / [CO1]

a). Dialogue Concerning the Two Chief World Systems

b). On the Revolutions of the Celestial Spheres

c). The Starry Messenger

d). Principia Mathematica

18. Name the two points where the celestial equator and the ecliptic intersects. [K] / [CO4]

a). Solstices b). Equinoxes

c). Zenith and Nadir d). Poles

19. What do you mean by Right Ascension? [K] / [CO4]

a). The celestial equivalent of Earth's Longitude

b). The celestial equivalent of Earth's Latitude

c). The celestial equivalent of Earth's North Pole

d). The celestial equivalent of Earth's South Pole

20. Hubble Space Telescope has primary mirror size of: [K] / [CO2]

a). 2.4 meters b). 2.4 kilometers

c). 10 meters d). 50 centimetres

21. How are neutron stars detected from Earth? [U] / [CO3]

a). By their visible light emissions b). By their gravitational waves

c). By their X-ray and radio wave emissions d). By their infrared radiation

22. What is the typical size and mass of a white dwarf? [U] / [CO3]

a). Same size as Jupiter with twice the Sun's mass

b). Same size as Earth with nearly the Sun’s mass

c). Larger than the Sun with half its mass

d). Same size as a comet with no significant mass

23. How many stars are there approximately in Milky Way galaxy? [K] / [CO4]

a). 200 billion b). 1 million

c). 10,000 d). 100,000

24. Which celestial body most commonly causes occultations seen from Earth? [U] / [CO5]

a). Moon b). Sun c). Mars d). Jupiter

25. What type of objects are found in the Kuiper Belt? [K] / [CO5]

a). Only large planets b). Icy bodies, dwarf planets, and comets

c). Gas clouds and stars d). Only meteors and asteroids

[2x10 = 20]

25901752 **B**

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Part B

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